

Container virtual networking is the software-defined infrastructure that allows isolated containers to communicate with each other, the host machine, and external networks 1, 2. In containerized environments like Docker, networking is handled by a software layer that abstracts physical hardware into virtual components 2.

1. Container Networking Concept

The core concept of container networking is the **Virtual Switch (vSwitch)**. In a virtualized or containerized environment, physical cables are replaced by virtual ones, and the hardware switch is replaced by a software-based vSwitch residing in the virtualization layer (hypervisor or container engine) 2. This vSwitch acts as a bridge between the containers and the physical network adapter (NIC) of the host 2, 3.

2. Network Types and Communication

- **Bridge Networks (Virtual Switching):** The sources describe the **vSwitch** as a bridge 2. When containers are on the same virtual switch, they can communicate using their **logical (Private IP) addresses** 4. The container engine intercepts network requests and ensures data is delivered to the correct container within this isolated segment 4.
- **Host Networking and Port Mapping:** Communication between the external world and the container happens through **Port Mapping** 5, 6. A specific **Host Port** (e.g., 8080) is mapped to a **Container Port** (e.g., 5000) 5. This allows the host to act as a gateway; traffic arriving at the host's public address is routed by the Docker Engine to the private IP of the container 5, 7.
- **Overlay and Global Networking:** While the sources do not explicitly define the term "Overlay," they describe **Global Server Load Balancing (GSLB)**, which routes traffic across different geographic regions and data centers to ensure users reach the nearest available virtual resource 8.

3. Communication Between Containers (ARP)

Containers use the **Address Resolution Protocol (ARP)** to map logical IP addresses to physical MAC addresses 9.

- **The Request:** When Container A wants to talk to Container B, it sends an ARP request broadcast: *"Which MAC address corresponds to this IP?"* 10.
- **The Interception:** In a virtual network, the **hypervisor or virtual networking software** intercepts this request 4, 11.
- **The Resolution:** If the destination is on the same vSwitch, the hypervisor provides the MAC address directly 4, 12. If the destination is on a different segment, a **virtual router or gateway** handles the resolution 12.

4. IP Addressing: Private vs. Public

- **Private IP (Internal):** These are like **student roll numbers** inside a college; they are meaningful only within the local container network and are not visible to the Internet 1, 3.

- **Public IP (External):** This is like the **official postal address** of the college. The entire host or network has one Public IP that identifies it on the Internet 3, 7.

Exam-Ready Explanation (10-Mark Answer)

I. Introduction

- Define container networking as the logical separation of traffic over shared physical infrastructure 7.
- Highlight the transition from physical hardware (switches/cables) to software-defined components (vSwitches/virtual cables) 2.

II. The Virtual Switch (vSwitch) Architecture

- Explain the role of the vSwitch as a **bridge** between the virtual machines/containers and the physical NIC 2, 3.
- Mention **Port Groups**, which are used to configure logical segments (VLANs) within the virtual switch to keep traffic isolated 2.

III. Addressing and Mapping

- Distinguish between **Private IPs** (internal identification) and **Public IPs** (external identification) 1, 7.
- Explain **ARP (Address Resolution Protocol)** as the mechanism for mapping logical IPs to permanent MAC addresses to ensure data delivery to the correct device 9, 13.

IV. Host-Container Interaction (Port Mapping)

- Detail the workflow: **Browser $\rightarrow$ Host Port $\rightarrow$ Docker Engine $\rightarrow$ Container Port $\rightarrow$ App Logic** 5.
- This ensures that isolated containers can still provide services to external users 5, 6.

V. Management Commands

- List essential Docker commands for networking: `docker network create` (creates a new network), `docker network connect` (links a container to a network), and `docker port` (shows port mappings) 14.

VI. Conclusion

- Summarize that virtual networking provides the **isolation, security, and scalability** required for modern applications while maximizing the use of a single physical host 7, 15.

Note: The specific technical definitions for "Host networks" and "Overlay networks" as formal Docker network drivers are not detailed in the provided sources, though their functional concepts (host interaction and global routing) are described.